

Biochemical basis of enhanced drug bioavailability by piperine: evidence that piperine is a potent inhibitor of drug metabolism.

Piperine, a major active component of black and long peppers, has been reported to enhance drug bioavailability. The present studies were aimed at understanding the interaction of piperine with enzymatic drug biotransforming reactions in hepatic tissue in vitro and in vivo. Piperine inhibited arylhydrocarbon hydroxylation, ethylmorphine-N-demethylation, 7-ethoxycoumarin-O-deethylation and 3-hydroxy-benzo(a)pyrene glucuronidation in rat postmitochondrial supernatant in vitro in a dose-dependent manner. Piperine inhibition of these reactions in postmitochondrial supernatant from 3-methylcholanthrene- and phenobarbital-treated rats was similar to the controls. Inhibition by piperine of arylhydrocarbon hydroxylase (AHH) from 3-methylcholanthrene-treated rats was comparable to that observed with 7,8-benzoflavone. Piperine caused noncompetitive inhibition of hepatic microsomal AHH from the untreated and 3-methylcholanthrene-treated rats with a K_i of 30 μM which was close to the apparent K_m of AHH observed in the controls. Similarly, the kinetics of inhibition of ethylmorphine-N-demethylase from control rat liver microsomes exhibited noncompetitive inhibition with an apparent K_m of 0.8 mM and K_i of 35 μM . These studies demonstrated that piperine is a nonspecific inhibitor of drug metabolism which shows little discrimination between different cytochrome P-450 forms. Oral administration of piperine in rats strongly inhibited the hepatic AHH and UDP-glucuronyltransferase activities. The maximal inhibition of AHH observed within 1 hr restored to normal value in 6 hr. Pretreatment with piperine prolonged hexobarbital sleeping time and zoxazolamine paralysis time in mice at half the dose of SKF-525A. These results demonstrate that piperine is a potent inhibitor of drug metabolism.